5.

$\xi \sim R[\theta, 2\theta]$ $(\theta > 0)$ $\theta \in \Theta = (0, +\infty)$

$p(x) = a\{[\theta, 2\theta]\}$

Найдем a:

$$\int_{\theta}^{2\theta} p(x)dx = 1 \Rightarrow a(2\theta - \theta) = 1 \Rightarrow a = \frac{1}{\theta}$$

$$p(x) = \frac{1}{\theta}\{[\theta, 2\theta]\}$$

$$M\xi = \int_{\theta}^{2\theta} \frac{1}{\theta} x\,dx = \frac{3}{2}\theta$$

$$M\xi^2 = \int_{\theta}^{2\theta} \frac{1}{\theta} x^2 dx = \frac{x^3}{3\theta} = \frac{7\theta^2}{3}; \quad D\xi = \frac{7\theta^2}{3} - \frac{9}{4}\theta^2 = \frac{1}{12}\theta^2$$

б) Несмещ. и сост.:

б.1) $\tilde{\theta}_1 = \frac{2}{3}\bar{x}$, $M(\tilde{\theta}_1) =$

$$D(\tilde{\theta}_1) = \frac{4}{9} \cdot \frac{1}{n} D\xi = \frac{4}{9} \frac{1}{n}$$

$$D\tilde{\theta}_1 = \frac{\theta^2}{27n}$$

а)

а.1) Метод моментов.

$$\alpha_1 = \int_{\theta}^{2\theta} x \cdot \frac{1}{\theta} dx = \frac{x^2}{2\theta}\Big|_{\theta}^{2\theta} = \frac{4\theta^2 - \theta^2}{2\theta} = \frac{3}{2}\theta$$

$\tilde{\alpha}_1 = \bar{x}$

$\alpha_1(\theta) = \tilde{\alpha}_1 \Rightarrow \tilde{\theta}_1 = \frac{2}{3}\bar{x}$

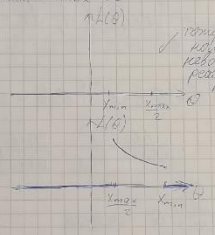
а.2) Метод макс. правдоподобия

$$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta) = \frac{1}{\theta^n}\{[\theta, 2\theta]\}$$

$$\theta \le x_{min} \le x_{max} \le 2\theta$$

Отр.

$$\sup_{\theta > 0} L(\theta) \longrightarrow \tilde{\theta}_2 = \frac{x_{max}}{2}$$



такое = вполне возможно, невозможная реализация нашей модели

б) Несмещ. и сост.:

б.1) $\tilde{\theta}_1 = \frac{2}{3}\bar{x}$, $M(\tilde{\theta}_1) = \frac{2}{3} \cdot M\xi = \frac{2}{3} \cdot \frac{3}{2}\theta = \theta \Rightarrow$ несмещенная

$$D(\tilde{\theta}_1) = \frac{4}{9} \cdot \frac{1}{n} D\xi = \frac{4}{9} \frac{1}{n} \cdot \frac{1}{12}\theta^2 \xrightarrow[n \to \infty]{} 0 \Rightarrow$$ по рассл. сост.

$$D\tilde{\theta}_1 = \frac{\theta^2}{27n}$$

б.2) $\tilde{\theta}_2 = \frac{x_{max}}{2}$;

$$x_{max} \sim \underbrace{F^n(x)}_{\varphi(x)}, \quad F(x) = R[\theta, 2\theta] = \begin{cases} \frac{x}{\theta} - 1, & x \in [\theta, 2\theta] \\ 0, & x < \theta \\ 1, & x > 2\theta \end{cases}$$

$$\int p(x)dx = \frac{x}{\theta} + C_{\text{на } [\theta, 2\theta]}$$

$$\frac{x}{\theta} + C\Big|_{\theta} = 0 \Rightarrow C = -1$$

$$\varphi(x) = \Phi'(x) = nF^{n-1}(x)p(x)_{[\theta, 2\theta]} = n \cdot \left(\frac{x}{\theta} - 1\right)^{n-1} \cdot \frac{1}{\theta}\{[\theta, 2\theta]\}$$

$$M x_{max} = \int_{\theta}^{2\theta} x \cdot \frac{n}{\theta}\left(\frac{x}{\theta} - 1\right)^{n-1} dx = \left\{\begin{array}{l} \frac{x}{\theta} - 1 = t \\ x = (t+1)\theta \end{array}\right\} =$$

$$= \int_0^1 (t+1)\theta \cdot \frac{n}{\theta} \cdot t^{n-1} \cdot \theta\,dt = n\theta \int_0^1 (t^n + t^{n-1})dt =$$

$$= n\theta\left(\frac{1}{n+1} + \frac{1}{n}\right) = \frac{2n+1}{n+1}\theta$$

$M(\tilde{\theta}_2) = \frac{2n+1}{2n+2}\theta$ – смещена

$\tilde{\theta}_3 = \frac{2n+2}{2n+1}\tilde{\theta}_2$ – несмещена

$D(\tilde{\theta}_3)$ – ?

$D(x_{max})$ – ?

$$M x_{max}^2 = \int_{\theta}^{2\theta} x^2 \cdot \frac{n}{\theta}\left(\frac{x}{\theta} - 1\right)^{n-1} dx =$$

$$M x_{max}^2 = \int_0^{2\theta} x^2 \frac{n}{\theta}\left(\frac{x}{\theta}-1\right)^{n-1} dx = \left\{ \begin{array}{l} \frac{x}{\theta}-1=t \\ x=(t+1)\theta \end{array} \right\} =$$

$$= \int_0^{1} (t+1)^2\theta^2 \frac{n}{\theta}\cdot t^{n-1}\cdot\theta\, dt = \theta^2 n\cdot\int_0^1 (t^{n+1}+2t^n+t^{n-1})\,dt =$$

$$= \theta^2 n\cdot\left(\frac{1}{n+2}+\frac{2}{n+1}+\frac{1}{n}\right) = \theta^2 n\left(\frac{n^2+n+2n^2+4n+(n+2)(n+1)}{(n+2)(n+1)n}\right)$$

$$= \theta^2\left(\frac{3n^2+5n+n^2+3n+2}{(n+2)(n+1)}\right) = \theta^2\left(\frac{4n^2+8n+2}{(n+2)(n+1)}\right)$$

$$D x_{max} = M x_{max}^2 - (M x_{max})^2 = \theta^2\frac{4n^2+8n+2}{(n+2)(n+1)} - \frac{(2n+1)^2}{(n+1)^2}\theta^2 =$$

$$= \frac{\theta^2}{n+1}\left(\frac{4n^3+8n^2+2n+4n^2+8n+2-4n^3-4n^2-n-8n^2-8n-2}{(n+2)(n+1)}\right) =$$

$$= \frac{\theta^2}{n+1}\left(\frac{n}{(n+2)(n+1)}\right) = \frac{n\theta^2}{(n+1)^2(n+2)}$$

$$D\tilde{\theta}_3 = \left(\frac{2n+2}{2n+1}\right)^2\cdot D\tilde{\theta}_2 = \left(\frac{2n+2}{2n+1}\right)^2\cdot\frac{1}{4}\cdot\frac{n\theta^2}{(n+1)^2(n+2)} = \frac{n\theta^2}{(2n+1)^2(n+2)} \xrightarrow{n\to\infty} 0 \Rightarrow$$

$\Rightarrow$ по дост. усл. состоятельн.

c) $D\tilde{\theta}_1 = \frac{\theta^2}{2n}$

$$D\tilde{\theta}_3 = \frac{n\theta^2}{(2n+1)^2(n+2)}$$

$D\tilde{\theta}_3 < D\tilde{\theta}_1$ при $n \geq 2$ $\Rightarrow$ $\tilde{\theta}_3$ эффективнее

d) $\xi \sim R[\theta, 2\theta]$

$\forall x_i \in [\theta, 2\theta] \Rightarrow \forall \frac{x_i}{\theta} \in [1, 2] \Rightarrow$

можно было сразу другую статистику

$\Rightarrow$ $\frac{X_{(n)}}{\theta}$

рассмотрим

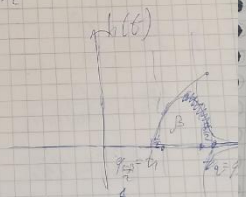
$$X_{(n)} \sim F(x) = (F(x))^n = \begin{cases} \left(\frac{x}{\theta} - 1\right)^n, & x \in [\theta, 2\theta] \\ 0, & x < \theta \\ 1, & x > 2\theta \end{cases}$$

$$y = \frac{X_{(n)}}{\theta} \sim H(t)$$

$$H(t) = P(y \le t) = P\left(\frac{X_{(n)}}{\theta} \le t\right) = P(X_{(n)} \le t\theta) =$$

$$= F(\theta t) = \begin{cases} \left(\frac{\theta t}{\theta} - 1\right)^n = (t-1)^n, & t \in [1, 2] \\ 0, & t < 1 \\ 1, & t > 2 \end{cases}$$

$$H'(t) = h(t) = \begin{cases} n(t-1)^{n-1}, & t \in [1, 2] \\ 0, & t \notin [1, 2] \end{cases}$$



Ищем квантили

$\beta = 0{,}95$

$H_{1-\frac{\alpha}{2}}(t_1) = \frac{\alpha}{2}$ $\quad (t_1 - 1)^n = \frac{\alpha}{2} \to t_1 = 1 + \left(\frac{\alpha}{2}\right)^{\frac{1}{n}}$

$H(t_2) = \beta + \frac{\alpha}{2}$ $\quad t_2 = 1 + \left(\beta + \frac{\alpha}{2}\right)^{\frac{1}{n}}$

$$P\left(t_1 < \frac{X_{(n)}}{\theta} < t_2\right) = \beta$$

$t_1 = 1 + \sqrt[n]{\frac{\alpha}{2}}$

$t_2 = 1 + \sqrt[n]{\beta + \frac{\alpha}{2}}$

$\frac{X_{(n)}}{t_2} < \theta < \frac{X_{(n)}}{t_1}$ — искомый доверительный интервал.

e) Асимптотический по ОМП.

из В.Г.Т.:

$$\frac{\sum x_i - n M\xi}{\sqrt{n D\xi}} \to N(0, 1)$$

$$\frac{n \cdot \tilde{\alpha}_1 - n\alpha_1}{\sqrt{n(\alpha_2 - \alpha_1^2)}} \to N(0, 1)$$

$$\sqrt{n}\,(\tilde{\alpha}_1 - \alpha_1) \to N(0, \alpha_2 - \alpha_1^2)$$

$\tilde{\theta}_1 = \frac{2}{3}\tilde{\alpha}_1$

$\theta = \frac{2}{3}\alpha$

$f(\alpha) = \frac{2}{3}\alpha$ — ~~дифф.~~ в т. α; $f'_\alpha = \frac{2}{3} \neq 0 \Rightarrow$

$\Rightarrow$ по $\tilde{\theta}$ послед. норм.

$$\sqrt{n} \cdot (\tilde{\theta}_1 - \theta_1) \to N\left(0, \frac{4}{9}(\alpha_2 - \alpha_1^2)\right)$$

$g(\theta) = \frac{2}{3}\sqrt{\alpha_2 - \alpha_1^2} = \frac{2}{3} \cdot \sqrt{D\xi} = \frac{2}{3} \cdot \frac{1}{2}\theta^2$ — непрер., $g(\theta) > 0$

$\tilde{\theta}_1 \xrightarrow{P} \theta$ (состоятельность)

$g(\tilde{\theta}_1) \xrightarrow{P} g(\theta)$, т.к. $g(x)$ — непрер.

$\downarrow$

$h(x) = \frac{\sigma(\theta)}{\sigma(x)}$ - непрер. на Θ $\Rightarrow$

$\Rightarrow h(\tilde{\theta}_1) \xrightarrow{P} h(\theta)$

$h(\tilde{\theta}_1) \xrightarrow{P} 1 \Rightarrow h(\tilde{\theta}_1) \xrightarrow{F} 1$

Из леммы Слуцкого:

$$\frac{\sqrt{n}(\tilde{\theta}_1 - \theta)}{\sigma(\theta)} \xrightarrow{F} N(0,1); \quad h(\tilde{\theta}_1) \xrightarrow{F} \overset{const}{1} \Rightarrow$$

$$\Rightarrow \frac{\sqrt{n}(\tilde{\theta}_1 - \theta)}{\sigma(\theta)} \cdot h(\tilde{\theta}_1) \xrightarrow{F} N(0,1)$$

$$\frac{\sqrt{n}(\tilde{\theta}_1 - \theta)}{\sigma(\tilde{\theta}_1)} \xrightarrow{F} N(0,1)$$

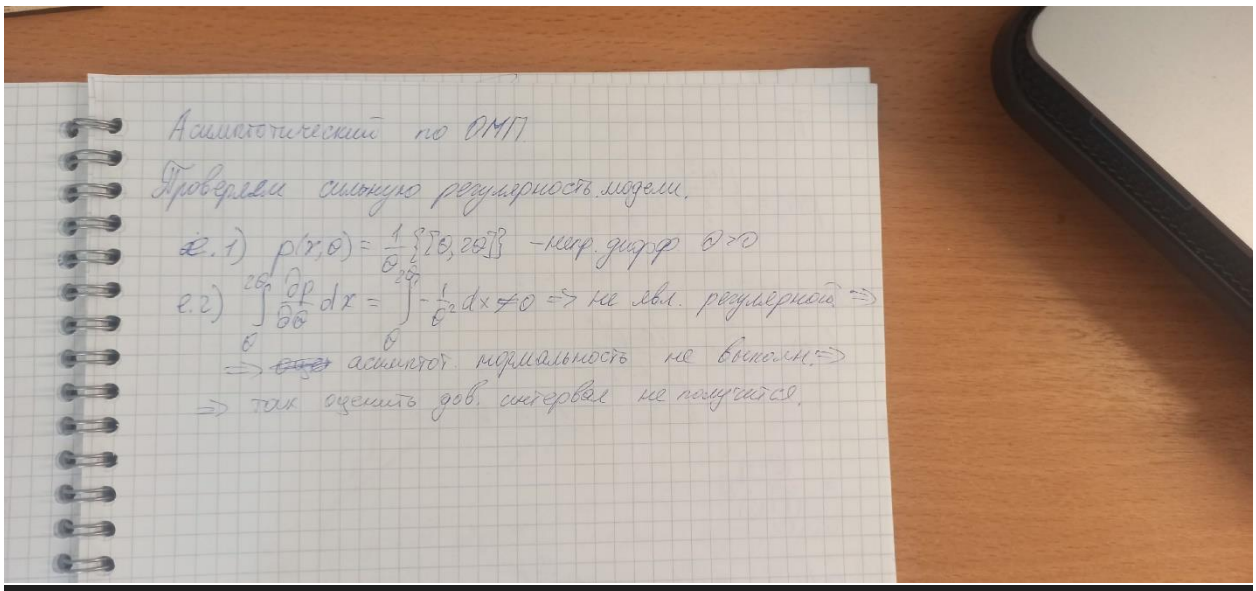
~~Тогда~~ Теперь можем найти доверит. интервал:

$t_1 = N_{\frac{1-\beta}{2}} \qquad t_2 = N_{\frac{1+\beta}{2}}$

$$t_1 < \frac{\sqrt{n}(\tilde{\theta}_1 - \theta)}{\sigma(\tilde{\theta}_1)} < t_2 \qquad \frac{t_1 \sigma(\tilde{\theta}_1)}{\sqrt{n}} - \tilde{\theta}_1 < -\theta < \frac{t_2 \sigma(\tilde{\theta}_1)}{\sqrt{n}} - \tilde{\theta}_1$$

$$\tilde{\theta}_1 - \frac{t_2 \sigma(\tilde{\theta}_1)}{\sqrt{n}} < \theta < \tilde{\theta}_1 - \frac{t_1 \sigma(\tilde{\theta}_1)}{\sqrt{n}}$$

$$\sigma(\tilde{\theta}_1) = \frac{2}{3} \cdot \sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}; \quad \tilde{\alpha}_2 = \frac{1}{n} \sum_{i=1}^{n} x_i^2; \quad \tilde{\alpha}_1 = \frac{1}{n} \sum_{i=1}^{n} x_i$$



Асимптотический по ОМП.

Проверяем сильную регулярность модели.

е.1) $p(x,\theta) = \frac{1}{\theta}\{[0, 2\theta]\}$ — непр. дифф $\theta > 0$

е.2) $\int_{\theta}^{2\theta} \frac{\partial p}{\partial \theta} dx = \int_{\theta}^{2\theta} -\frac{1}{\theta^2} dx \neq 0 \Rightarrow$ не явл. регулярной $\Rightarrow$

$\Rightarrow$ ~~отсю~~ асимптот. нормальность не выполн. $\Rightarrow$

$\Rightarrow$ так оценить дов. интервал не получится.

```
teta = 10
точный доверительный интервал: 9.958279912378174 < teta < 10.140649862496822
его длина: 0.1823699501186482
```

```
асимптотический доверительный интервал по ОММ: 9.554683184031745 < teta < 10.274750437105354
его длина: 0.7200672530736085
```

```
бустраповский (непараметрический) доверительный интервал по ОММ: 9.540960597865482 < teta < 10.274810082819712
его длина: 0.7338494849542307
```

```
бустраповский (параметрический) доверительный интервал по ОММ: 9.538604718397963 < teta < 10.2887383078086
его длина: 0.7501335894106376
```

```
бустраповский (непараметрический) доверительный интервал по ОМП: 10.006556873113563 < teta < 10.157137337595863
его длина: 0.15058046448229945
```

```
бустраповский (параметрический) доверительный интервал по ОМП: 9.866240562541766 < teta < 10.050203018958289
его длина: 0.18396245641652342
```